

Unsupervised Discovery of Urban Mobility Archetypes from High-Volume NYC Taxi Trip Records

MSc Big Data Analytics — Research Proposal

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Programme: MSc Big Data Analytics

Research Theme: Urban Mobility & Transport Analytics

Dataset: NYC Taxi & Limousine Commission, Yellow Taxi Trip Records (2023, Jan–Jun)

Scale: ~19.5 million trips · 263 zones · 6 × 50 MB Parquet files (~305 MB compressed Parquet; >2.5 GB uncompressed in memory)

1. Motivation and Research Gap

Urban transportation systems generate some of the densest, most temporally rich open datasets available to researchers. A single month of New York City yellow taxi records encodes millions of individual mobility decisions: where people travel, when they travel, how much they pay, and which spatial corridors they inhabit. Yet despite this richness, most published taxi analytics often falls into one of two dominant tracks: supervised demand forecasting (predicting trip counts given historical time series) or descriptive cartography (heat maps of pickup density).

Neither approach exploits the *latent structure* of the data. Neither asks: **what archetypes of spatial demand exist across the city's 263 zones?** Or: **do zones share temporal demand rhythms independent of their absolute volume?** Or: **which individual trips are structurally unusual in ways that signal regulatory concern?**

These are genuinely unsupervised questions. They require no labels, no ground-truth outcomes, and no predefined hypotheses about which zones are "similar" or which hours are "important." They are exactly the kind of questions that Big Data analytical frameworks, applied rigorously at scale, are positioned to answer.

This proposal situates a six-layer unsupervised learning pipeline within the Big Data 5 Vs framework and argues that NYC taxi trip records represent a model case study for demonstrating the analytical value of high-volume, regularly generated urban mobility data.

NYC TLC data is selected because it is one of the few fully open, high-volume, trip-level urban mobility datasets with consistent spatial, temporal, and financial variables. While the empirical setting is New York City, the methodology is transferable to Ghanaian urban transport contexts where comparable data may become available through digital ticketing, ride-hailing platforms, GPS fleet tracking, or public transport smart-card systems. This proposal therefore serves both as a standalone analytical contribution and as a methodological template for future work on African urban mobility data.

2. Research Questions

RQ1 (Spatial): What demand archetypes emerge when NYC taxi zones are characterised by their aggregate volume, revenue, behavioural, and temporal profiles, and do these archetypes correspond to meaningful urban functional categories?

RQ2 (Temporal): Do taxi zones share demand rhythm *shapes* independent of absolute trip volume, and if so, what distinct rhythm archetypes can be identified by shape-based time-series clustering?

RQ3 (Structural): What latent spatiotemporal demand components can be extracted from the Zone $\times$ Hour $\times$ DayType demand tensor, and how many independent archetype components are required to explain the majority of observed variation?

RQ4 (Anomaly): Which individual trips are structurally unusual relative to the distribution of comparable trips, and which zones exhibit the highest concentration of anomalous behaviour?

RQ5 (Synthesis): Does a unified multi-layer zone fingerprint combining spatial, temporal, tensor, and anomaly signals produce a coherent low-dimensional embedding that reflects the functional geography of New York City?

3. Dataset and Big Data 5 Vs Assessment

Data will be sourced from the NYC Taxi & Limousine Commission public Yellow Taxi Trip Records for January–June 2023, retrieved from the TLC open data portal in Parquet format.

Dimension	Assessment	Evidence
Volume	★★★★★	19.5M raw trips in 6 months (18.8M after filtering); 305 MB Parquet; ~100M trips/year system-wide
Velocity	★★★★☆	Trip data is generated continuously by taxi operations; public research access is through periodic monthly batch releases
Variety	★★★★☆	Spatial (zone IDs, coordinates), temporal (datetime), financial (fare, tip, surcharges), behavioural (passenger count, payment type, ratecode)
Veracity	★★★☆☆	Known GPS dropouts, metering anomalies, zone boundary edge cases; quality filtering required
Value	★★★★★	Direct policy relevance: TLC regulation, congestion pricing, accessibility equity, compliance review

The dataset strongly satisfies the Volume, Variety, Veracity, and Value dimensions. Velocity is represented through the high frequency of operational data generation and regular public release cadence, even though this study uses batch processing rather than streaming analysis. The combination of spatial resolution (263 named zones, many mappable to census tracts), temporal granularity (second-level timestamps), and scale (tens of millions of records per year) makes it qualitatively different from datasets processable by conventional analytical tools, justifying the Big Data framing.

4. Analytical Framework: Multi-Stage Unsupervised Pipeline

The analysis proposes a rigorous sequential multi-method unsupervised framework. Individual methods (K-Means spatial clustering, K-Shape time-series clustering, Tucker tensor decomposition, Isolation Forest anomaly detection, and UMAP embedding) each have established use in their respective literatures, but their combination into a single coherent pipeline applied to one dataset is not a standard approach in the urban mobility field. The core contribution is the integrated zone fingerprint: K-Means, K-Shape, Tucker decomposition, and anomaly rates are intermediate feature-generation layers, while UMAP provides the final synthesis across all dimensions. The outputs of earlier stages feed into later ones through this

unified fingerprint, producing a richer characterisation than any single method could achieve alone.

Layer 1 — Data Loading, Quality Filtering, Feature Engineering

Method: Pandas + PyArrow (Parquet loading); domain-knowledge quality filter.

Quality constraints applied: - Duration: 1–180 minutes - Distance: 0.1–60 miles - Fare: \$2.50–\$500 - Speed: ≤ 80 mph - Zone: valid TLC zone IDs (1–263)

Features engineered: `duration_min`, `speed_mph`, `tip_pct`, `hour`, `dow`, `is_weekend`, `month`. Preliminary results indicate approximately 98.5% of raw trips are retained after filtering, suggesting the dataset has high intrinsic quality.

Big Data justification: Parquet columnar storage enables selective column loading without materialising the full 300+ MB dataset in memory. Filtering at this scale requires vectorised Pandas operations rather than row-by-row processing.

Layer 2 — K-Means Spatial Clustering: Demand Archetypes

Method: K-Means on 10-feature zone profile matrix; elbow + silhouette selection of k ; violin plot portraits per cluster.

Features: `log(trip_count)`, `mean_fare`, `mean_distance`, `mean_duration`, `mean_speed`, `tip_pct`, `pct_weekend`, `pct_cash`, `pct_night`, `pct_rush`. All zone-level features will be standardised using z-score scaling before K-Means clustering to prevent high-magnitude variables from dominating distance calculations. Sensitivity checks will be conducted with and without airport zones to ensure that large hub effects do not distort the broader cluster structure.

Preliminary clustering results ($k=5$ by silhouette, based on 19.5M trips):

Cluster	Character	Top zones	Avg fare	Avg dist	% night
0	High-volume Midtown/UES	Upper East Side South, Midtown Center	\$16.94	2.85 mi	15%
1	Outer-borough residential	Crown Heights N, Clinton Hill, Bedford	\$30.89	6.73 mi	13%

Cluster	Character	Top zones	Avg fare	Avg dist	% night
2	Nightlife/entertainment	East Village, West Village, Greenwich Vill.	\$23.34	4.55 mi	49%
3	Underserved communities	East Harlem N, Chinatown, Central Harlem N	\$24.87	4.61 mi	19% · 31% cash
4	Airport/Hub	JFK Airport, LaGuardia Airport	\$46.93	10.23 mi	16%

The 49% night-trip share in Cluster 2 is the sharpest behavioural signal in the preliminary data; nightlife zones appear to operate on a fundamentally different temporal regime from all others. Cluster 3's 31% cash share may indicate differences in payment behaviour or digital payment access, requiring cautious interpretation before causal claims are made.

This layer is designed to answer **RQ1** and will establish the zone typology used throughout the analysis.

Layer 3 — K-Shape: Temporal Rhythm Clustering

Method: K-Shape clustering on normalised 24-hour demand profiles per zone; silhouette-selected k. To prevent sparsity-induced shape distortion, zones averaging fewer than 10 trips per day are excluded, leaving 121 highly active zones for time-series analysis. K-Shape uses cross-correlation-based shape distance, which is computationally efficient at this scale. DTW K-Means is also run for comparison; K-Shape results are reported as primary given superior silhouette scores.

Innovation: Normalising demand profiles to *fractional hourly share* before clustering decouples rhythm shape from volume. A low-volume residential zone that peaks at 8 AM will cluster with Midtown, which also peaks at 8 AM, despite a 100-fold volume difference. This is intentional: the research question is about *rhythm archetype*, not scale.

Preliminary rhythm archetypes (k=2 by silhouette, 121 active zones):

Archetype	Borough mix	Character
Rhythm 0	10 Brooklyn + 11 Manhattan + 3 Queens	Midday-dominant, moderate peaks

Archetype	Borough mix	Character
Rhythm 1	4 Bronx + 19 Brooklyn + 49 Manhattan + 25 Queens	Business-hour dominant; dense Manhattan core

The two-archetype solution appears statistically robust in preliminary analysis ($k=2$ maximises silhouette). Initial results suggest that functional zone type may explain demand shape more strongly than simple geographic proximity, though this interpretation will be validated against the full analysis.

Findings from Layer 3 will therefore be interpreted as rhythm archetypes among active taxi zones, not all 263 TLC zones.

This layer is designed to answer **RQ2**.

Layer 4 — Non-negative Tucker Decomposition

Method: Non-negative Tucker decomposition of the Zone $\times$ Hour $\times$ DayType ($Z \times 24 \times 2$) demand tensor; core tensor rank set to (6, 4, 2): 6 latent zone profiles, 4 temporal profiles, and 2 day-type profiles. This low-rank decomposition meaningfully compresses the $263 \times 24 \times 2$ tensor (12,624 cells) into interpretable latent components. Rank selection will be justified using reconstruction error, component interpretability, and sensitivity checks against nearby ranks such as (4,3,2), (5,4,2), and (8,4,2). Since DayType has only two categories, rank 2 preserves the full weekday/weekend distinction while allowing meaningful compression in the zone and hour modes.

Rationale for Tucker over standard matrix factorisation: The demand tensor is three-dimensional by construction: a zone's demand is a function of *both* the hour of day *and* the day type (weekday vs weekend). A matrix factorisation that collapses one dimension loses this interaction structure. Tucker NTD preserves all three modes simultaneously, yielding: - Zone factors: how much each zone participates in each latent archetype - Hour factors: the temporal profile of each archetype - Day-type factors: how archetypes modulate across weekday/weekend

Non-negativity is enforced because demand cannot be negative, which also improves interpretability of the component loadings.

This layer is designed to answer **RQ3**.

Layer 5 — Isolation Forest: Trip-Level Anomaly Detection

Method: Isolation Forest (200 trees, 2% contamination) on a 500,000-trip random sample; 7-feature trip-level input matrix.

Features: trip_distance, duration_min, fare_amount, speed_mph, tip_pct, hour, is_weekend.

Preliminary results (500k-trip sample, 2% contamination):

- 10,000 trips flagged as anomalous (2.0%)
- Median anomaly fare: **\$70.00** vs normal **\$13.50** (5.2× higher)
- Median anomaly speed: **31.0 mph** vs normal **9.8 mph** (3.2× higher)
- Median anomaly distance: 19.05 mi vs normal 1.80 mi

This profile (high speed, high fare, long distance) is consistent with trips that warrant further regulatory review, such as unusual fare-distance-speed combinations. The anomaly layer is not designed to prove fraud; it identifies unusual trip structures that may support further compliance review by the relevant authority. Staten Island zones appear at the top of the anomaly-rate list (Arrochar/Fort Wadsworth: 100%, South Beach: 100%). This is a known statistical artefact caused by sparse observations: Yellow Taxis almost never serve Staten Island, so these zones recorded fewer than 5 pickups over the full six-month period. A single unusual trip in a near-zero-volume zone produces a 100% anomaly rate by construction. This is treated as a data sparsity effect, not a signal of systematic irregularity in those locations.

The per-zone anomaly rate will serve as the final feature in the UMAP fingerprint and as a candidate regulatory signal for zones with persistently elevated rates.

This layer is designed to answer **RQ4**.

Layer 6 — UMAP: Unified Zone Fingerprint

Method: UMAP (n_neighbors=10, min_dist=0.25) on 17-dimensional zone fingerprint; coloured by Layer 2 and Layer 3 cluster membership.

Fingerprint composition: Layer 2 zone features (10) + Tucker zone factors (6) + anomaly rate (1) = 17 dimensions per zone.

Preliminary or expected embedding structure: - Manhattan zones: dense, well-separated cluster reflecting high-volume homogeneity - Airport zones: isolated outliers with unique multi-

modal demand profiles - Brooklyn/Queens: partial overlap reflecting similar residential rhythms
- Bronx: peripheral region with outer-borough character

UMAP will be used as an exploratory visualisation tool, not as a confirmatory statistical test. This layer is designed to answer **RQ5** and will provide the holistic synthesis figure for the dissertation.

5. Method Validation Strategy

Each unsupervised layer will be evaluated using the following criteria to ensure outputs are analytically defensible rather than artefacts of parameter choices.

Layer	Validation Approach
K-Means (Layer 2)	Elbow method, silhouette score, cluster profile interpretability against known NYC geography
K-Shape / DTW (Layer 3)	Silhouette score, centroid inspection, comparison of k=2 through k=6; sensitivity to zone exclusion threshold
Tucker NTD (Layer 4)	Reconstruction error across rank settings (4,3,2) through (8,6,2); component interpretability; rank sensitivity analysis
Isolation Forest (Layer 5)	Stability of zone-level anomaly rates across independent random samples; contamination parameter sensitivity at 1%, 2%, and 5%
UMAP (Layer 6)	Neighbourhood preservation/trustworthiness score; visual separation aligned with known borough/zone categories; comparison with PCA as baseline

6. Theoretical Grounding

6.1 Urban Informatics and the Smart City Paradigm

This work sits within the emerging field of *urban informatics*, the application of computational methods to the dense data streams produced by modern cities (Batty, 2013). Taxi trip records are a canonical urban informatics dataset: they encode individual mobility behaviour at city-scale,

without the privacy costs of GPS tracking or mobile data, because the TLC mandates anonymised public release.

The Big Data 5 Vs framework (Laney, 2001) provides the analytical scaffolding: Volume justifies distributed processing; Variety motivates multi-modal feature engineering; Velocity contextualises real-time policy applications; Veracity drives the quality filtering layer; Value grounds the regulatory and equity implications.

6.2 Unsupervised Learning as Hypothesis Generation

Unlike supervised models, which require labelled outcomes and test pre-specified hypotheses, unsupervised learning treats the data as the primary source of structure. This is epistemologically appropriate for exploratory urban analytics, where the goal is not to test a known hypothesis but to discover which hypotheses are worth testing.

The pipeline therefore follows the *discovery* → *characterisation* → *validation* cycle advocated by Tukey (1977) in exploratory data analysis, extended to the machine learning era by Hastie et al. (2009). Each layer generates a finding that becomes a candidate for confirmatory analysis in future work: the cluster assignments become treatment groups for regression; the anomaly flags become inputs for supervised compliance classifiers; the UMAP embedding becomes the basis for spatial econometric modelling.

6.3 Tensor Methods in Urban Analytics

The application of Tucker decomposition to urban mobility tensors is methodologically grounded in the growing literature on tensor-based spatiotemporal analysis (Kolda & Bader, 2009; Cichocki et al., 2016). Kolda & Bader (2009) provide the mathematical foundation for both Tucker and CP decompositions, while Cichocki et al. (2016) establish non-negativity constraints as a route to interpretable components. This analysis applies those foundations to a public, reproducible urban mobility dataset, making the methodology accessible for replication and extension.

7. Research Plan

Phase 1 — Data Infrastructure (Weeks 1–2)

Task	Deliverable
Download and validate 2023 TLC Parquet files (Jan–Jun)	<code>data/raw/nyc-taxi-trips/</code> (~305 MB)
Implement quality filter pipeline	Documented filter script
Engineer temporal/spatial features	Feature matrix: $\sim 19.5\text{M} \times 15$ columns
Profile data against 5 Vs criteria	Appendix table in dissertation

Phase 2 — Core Analysis (Weeks 3–6)

Task	Deliverable
Layer 2: Zone K-Means clustering (k selection, portraits)	Figure: violin plots per cluster
Layer 3: K-Shape rhythm clustering	Figure: centroid rhythm archetypes
Layer 4: Tucker NTD decomposition	Figure: hourly component loadings + zone scatter
Layer 5: Isolation Forest anomaly detection	Figure: anomaly scatter plots + zone heatmap
Layer 6: UMAP fingerprint embedding	Figure: 2-D zone map with borough labels

All outputs saved to `data/processed/` and `reports/figures/` for reproducibility.

Phase 3 — Synthesis and Writing (Weeks 7–8)

Task	Deliverable
Produce dissertation chapter draft	Chapter document
Cross-reference findings to 5 Vs framework	Section 2 of chapter
Document pipeline design decisions and layer outputs	Methods section of chapter
Peer review of findings with supervisor	Revision notes

Phase 4 — Extension (Weeks 9–10, optional)

Extension	Justification
Incorporate 2020 data (COVID impact)	Strong natural experiment for mobility disruption
Add green taxi and FHV (Uber/Lyft) data	Tri-modal comparison; market segmentation analysis
Implement DBSCAN spatial clustering	Density-based zones handle irregular NYC geography better than K-Means

8. Expected Contributions

8.1 Empirical Contributions

1. **Zone typology:** A reproducible five-cluster taxonomy of NYC taxi demand archetypes, derived from 19.5M trips and validated against known urban geography.
2. **Rhythm atlas:** A borough-level characterisation of temporal demand rhythms, with preliminary evidence that functional zone type may explain demand shape more strongly than geographic proximity.
3. **Anomaly geography:** A zone-level map of structurally unusual trip concentrations, quantifying the spatial distribution of fare-distance-speed irregularities at city scale.
4. **Tensor decomposition of urban demand:** Evidence that the Zone $\times$ Hour $\times$ DayType demand tensor may admit a low-rank non-negative factorisation, suggesting NYC taxi demand could be governed by a small number of latent mobility archetypes.

8.2 Methodological Contributions

1. **Multi-layer pipeline integration:** The six-layer framework (data profiling $\rightarrow$ K-Means $\rightarrow$ K-Shape $\rightarrow$ Tucker NTD $\rightarrow$ Isolation Forest $\rightarrow$ UMAP) combines methods from distinct analytical traditions into a single coherent pipeline. Layers 1–4 build the zone profile sequentially; Layer 5 (Isolation Forest) operates in parallel on the raw filtered trip sample to compute zone-level anomaly rates, which are then joined as the final feature in the 17-dimensional fingerprint fed to Layer 6 (UMAP). This design demonstrates that

spatiotemporal structure in urban mobility data can be characterised simultaneously across spatial, temporal, tensor, anomaly, and embedding dimensions.

2. **Scale demonstration:** Core zone-level aggregation uses the full filtered dataset of 18.8M trips. K-Shape is applied to the subset of sufficiently active zones (121 of 263) to avoid sparse-profile distortion, while Isolation Forest uses a 500k representative sample for computational tractability. Together, the pipeline demonstrates operation at genuine Big Data scale without requiring cloud infrastructure.
3. **Open reproducibility:** The full pipeline, from raw Parquet download through final figures, is encapsulated in a single Jupyter notebook executable from the project venv.

8.3 Policy Contributions

1. **Regulatory targeting:** Zone-level anomaly rates provide a ranked list of zones for TLC compliance review, which is more informative than random auditing.
2. **Infrastructure planning:** Zone demand archetypes and rhythm clusters inform bus route redesign and bike-share station placement, which depend on peak demand timing rather than daily totals.
3. **Equity lens:** The spatial distribution of zone clusters, and whether outer-borough residential zones are systematically underserved by high-fare, credit-card-dominant taxi service, has direct implications for transportation equity policy.

9. Relationship to Existing Literature

Reference	Relevance
Yuan et al. (2011) — passenger hotspots	Demand hotspot prediction for taxi drivers; motivates zone-level demand clustering
Castro et al. (2012) — taxi GPS modelling	Traffic modelling and prediction from large-scale taxi GPS traces; methodological context
Zheng et al. (2014) — Urban computing survey	Broad review of urban computing methods including spatiotemporal data analysis

Reference	Relevance
Colak et al. (2016) — urban congestion	Mobile phone-based congestion modelling; motivates multi-modal mobility analysis
Laney (2001) — 5 Vs	Big Data framing and assessment criteria
Kolda & Bader (2009) — Tensor decompositions	Mathematical foundation for Tucker and CP decompositions
Paparrizos & Gravano (2015) — K-Shape	Shape-based time-series clustering algorithm used in Layer 3
McInnes et al. (2018) — UMAP	Dimensionality reduction algorithm used in Layer 6 for zone fingerprint embedding

10. Limitations and Mitigations

Limitation	Mitigation
Data covers Jan–Jun only — seasonal bias	Flag explicitly; propose full-year extension in future work
Taxi market share declining vs FHV (Uber/Lyft)	Acknowledge selection bias; yellow taxi data is still the only fully open, trip-level dataset
Zone-level analysis loses within-zone heterogeneity	Layer 5 (trip-level anomaly) partially recovers individual-trip structure
Preliminary Tucker reconstruction error leaves 32% unexplained variation	Human mobility tensors are inherently noisy; 68% explained variance captures structural macro-trends while omitting high-frequency transactional noise. A higher rank risks overfitting given the tensor sparsity ($263 \text{ zones} \times 24 \text{ hours} \times 2 \text{ day types} = 12,624 \text{ cells}$).
IsoForest trained on 500k sample, not full 19.5M	Validate: train on random halves, check that anomaly rates per zone are stable

11. Data Ethics and Privacy

NYC TLC trip records are published as open government data under the NYC Open Data program. The dataset is anonymised at source: no driver or passenger identifiers are included. Pickup and dropoff locations are reported at zone level (not GPS coordinates), preventing re-identification of individual trips. No personally identifiable information is processed or stored at any stage of this analysis.

The anomaly detection layer identifies *trip-level* patterns, not individual actors. Any regulatory application of anomaly scores would require separate data linkage by TLC under their existing compliance mandate; this research produces only aggregate zone-level anomaly rate statistics.

12. References

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